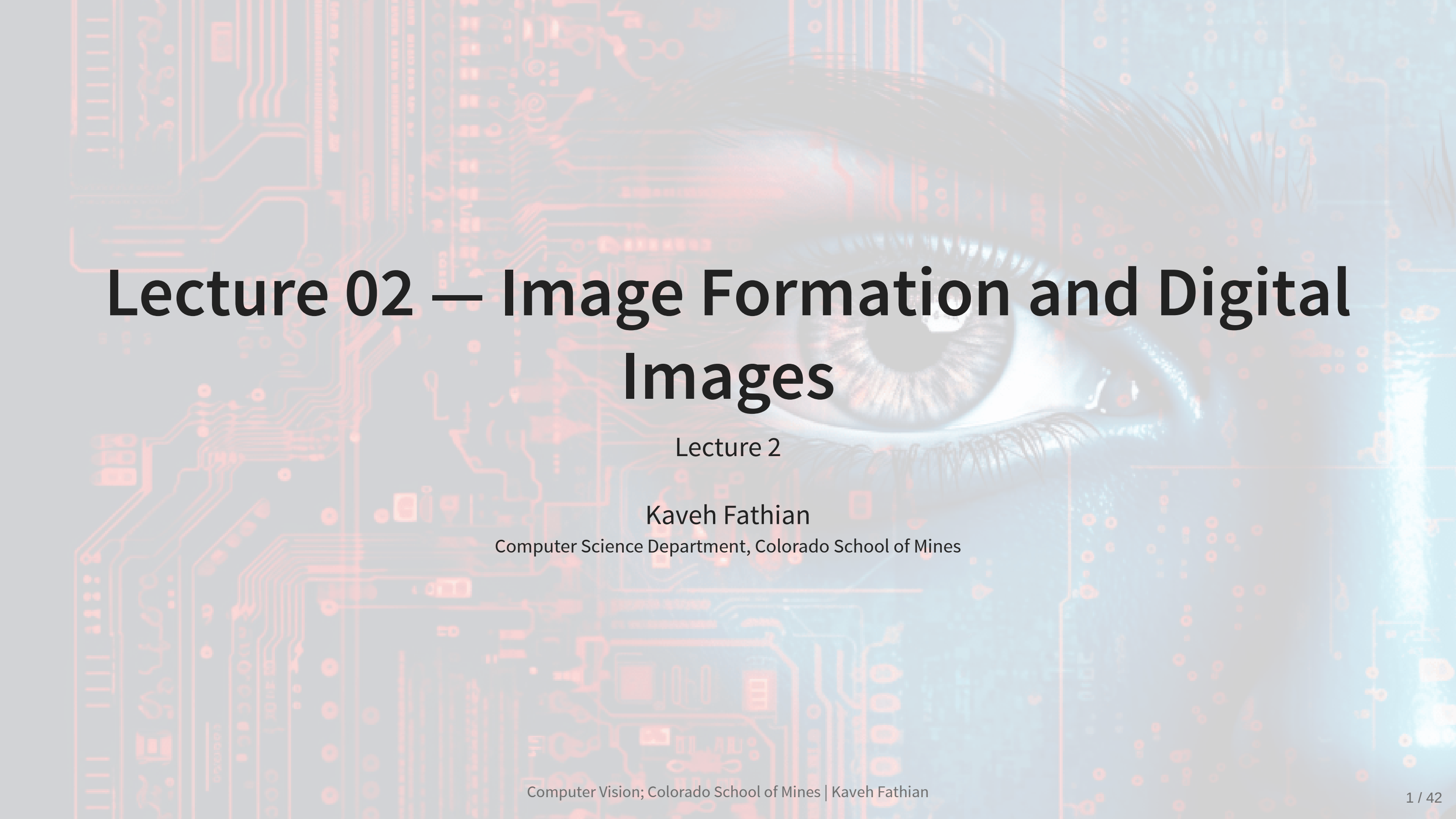




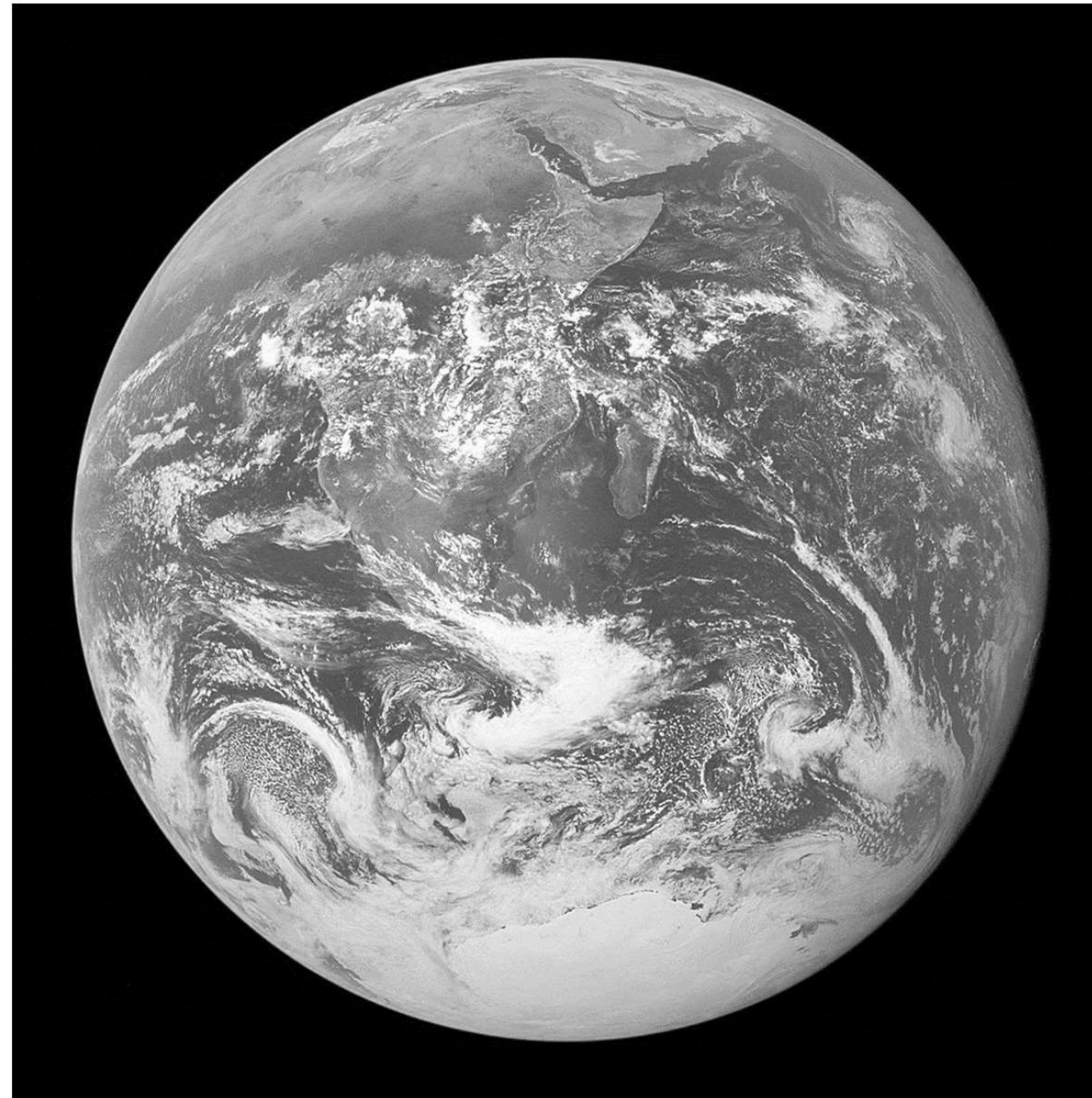
Lecture 02 — Image Formation and Digital Images

Lecture 2

Kaveh Fathian

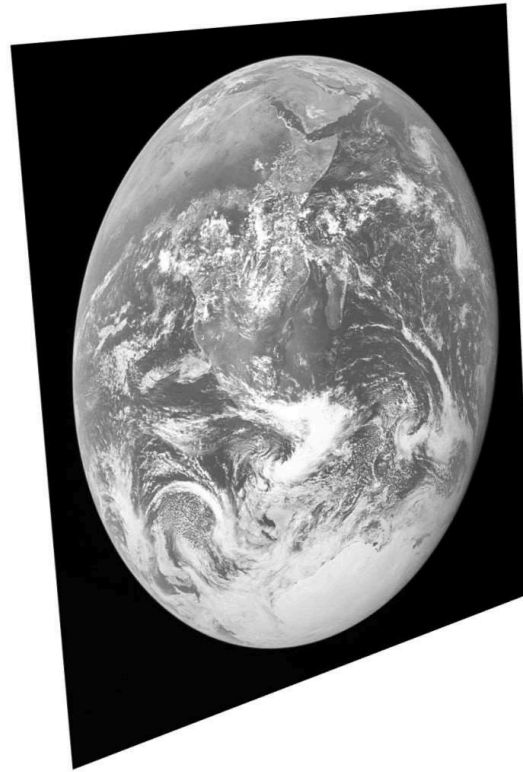
Computer Science Department, Colorado School of Mines

What is an Image and how is it formed?

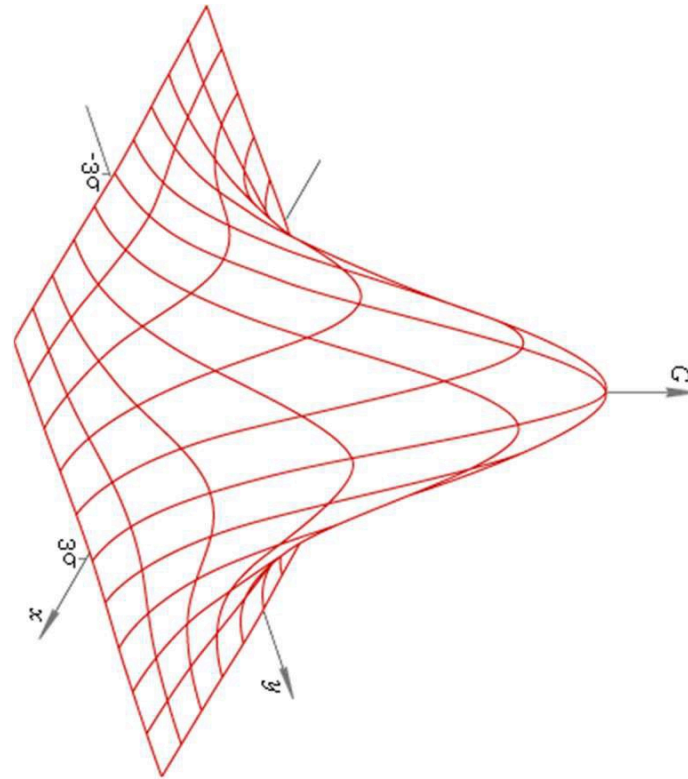


Blue Marble, NASA / Apollo 17

Image Formation



Scene / object



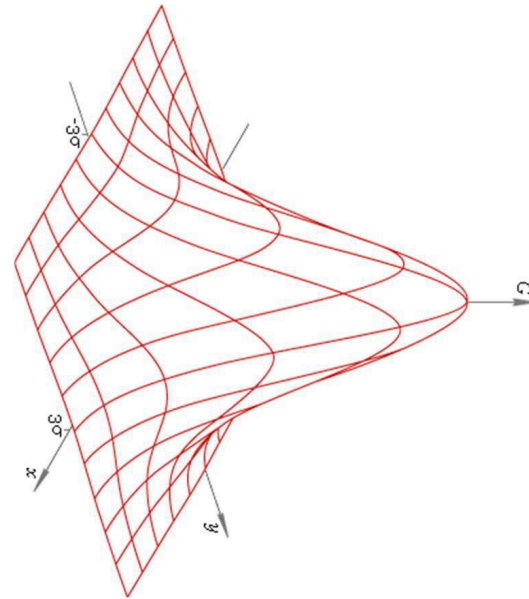
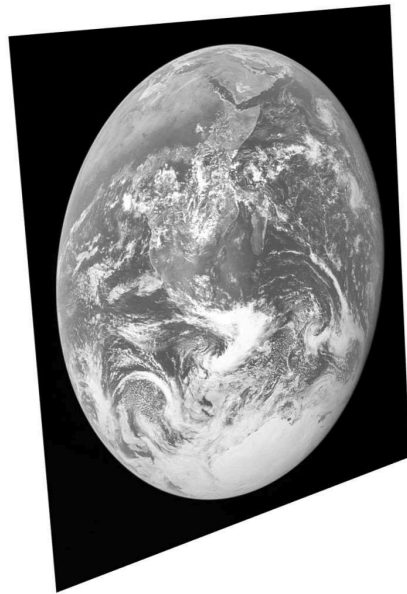
Light as a signal



Camera / sensor

A camera measures light from the world and turns it into a digital image.

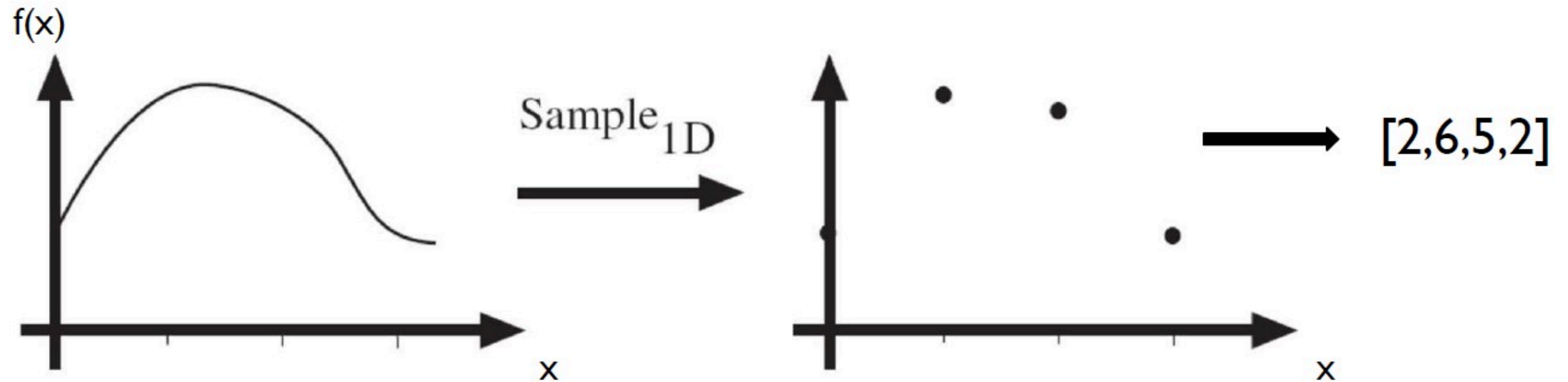
Signal



Signal: A (multi-dimensional) function that contains information about a phenomenon (e.g., light, heat, gravity, sound, pressure, motion).

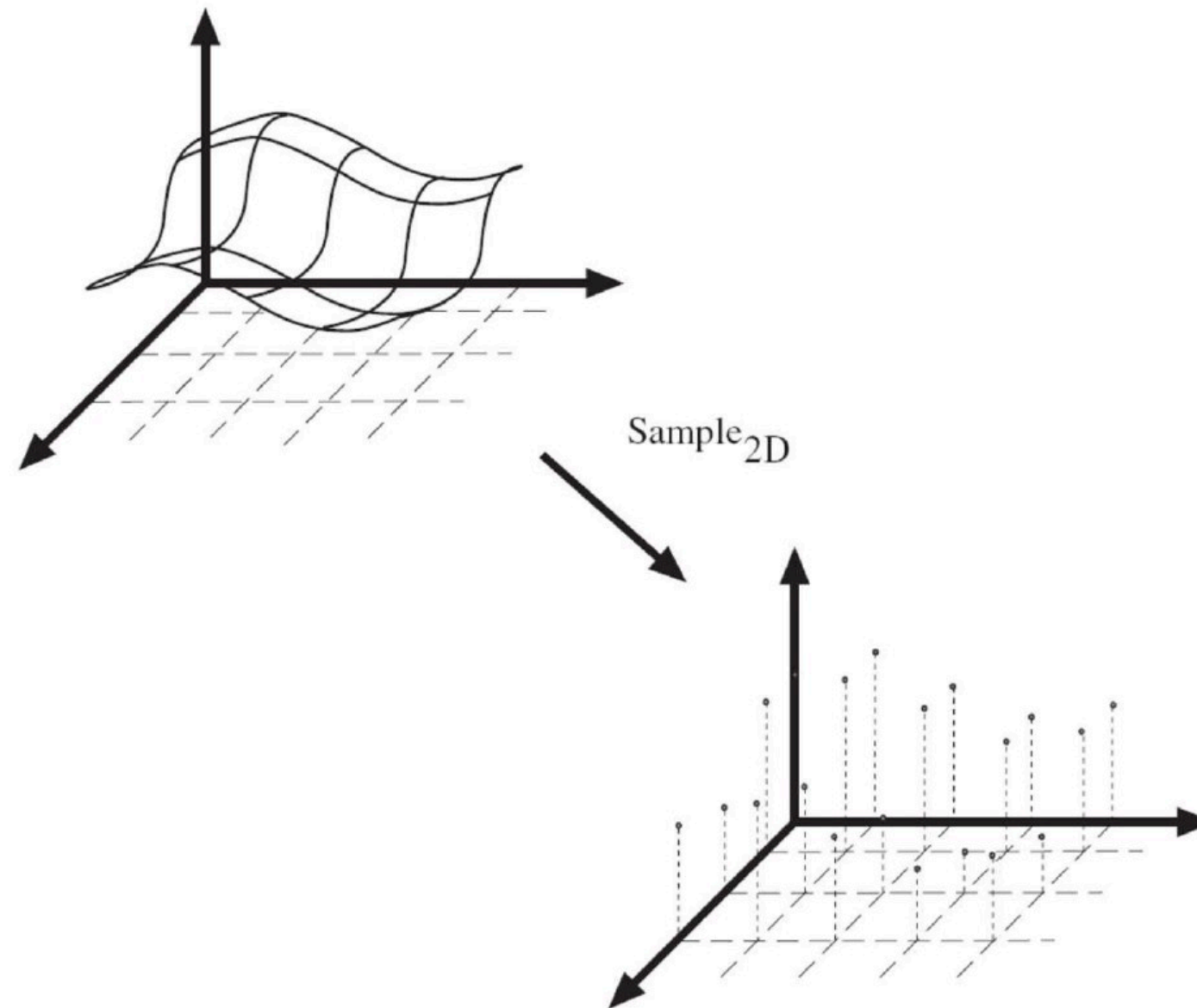
- Signals can be **continuous**, such as light in the physical world.
- Measurements are often **discrete** samples of that continuous signal.
- Sampling reduces a continuous signal to a finite set of values.

Sampling in 1D



- Sampling a 1D continuous function returns a vector whose elements are values of that function at selected sample points.
- Instead of the full function, we store values at selected locations.

Sampling in 2D



Sampling a 2D function returns a matrix.

2D Image

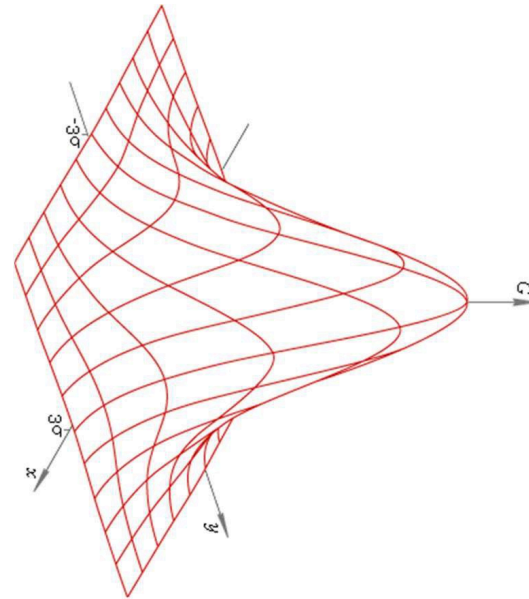
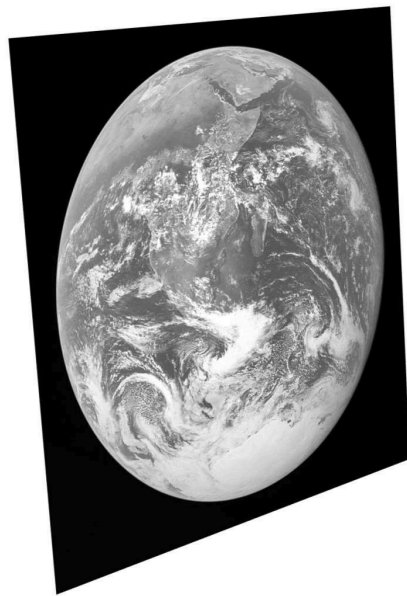


Image: A sampling of a function that contains information about a 2D signal.

- The function stores **brightness** or **intensity**.
- The image samples brightness along the x and y dimensions.
- A video can be viewed as a time-varying 2D image signal: (x, y, t) .

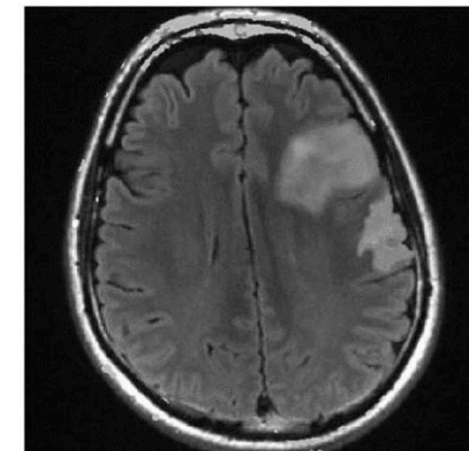
i Note

Cameras observe a 2D projection of the 3D world.

Examples of 2D Images

2D images appear in many sensing modalities:

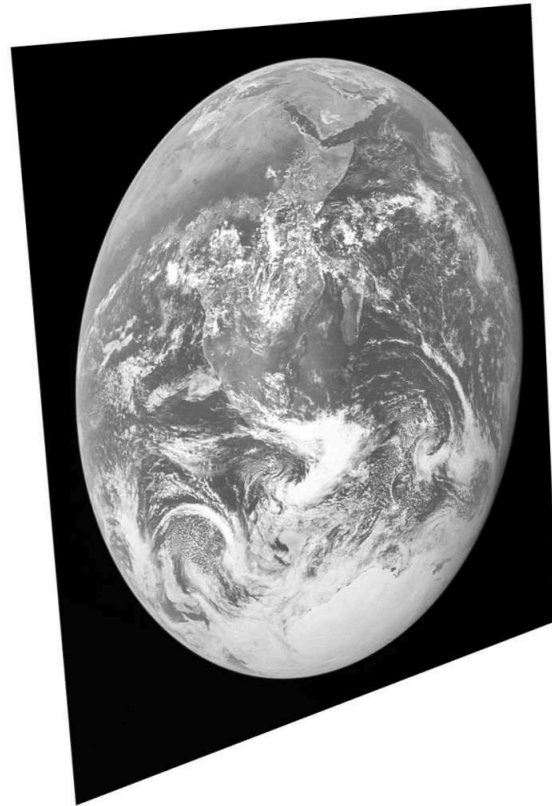
- Natural images
- Thermal images
- Satellite/aerial images
- Microscopy
- Ultrasound
- MRI / CT slices
- Radar images
- Depth images



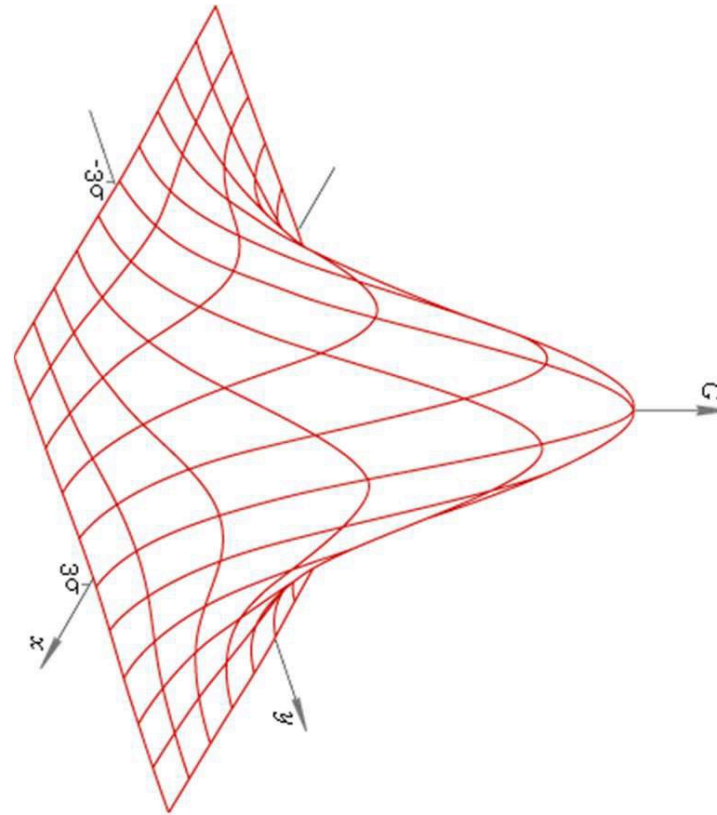
i Note

The signal may **not** always be visible light. An image arranges what we have measured over a 2D domain.

Sampling in Practice



World / scene



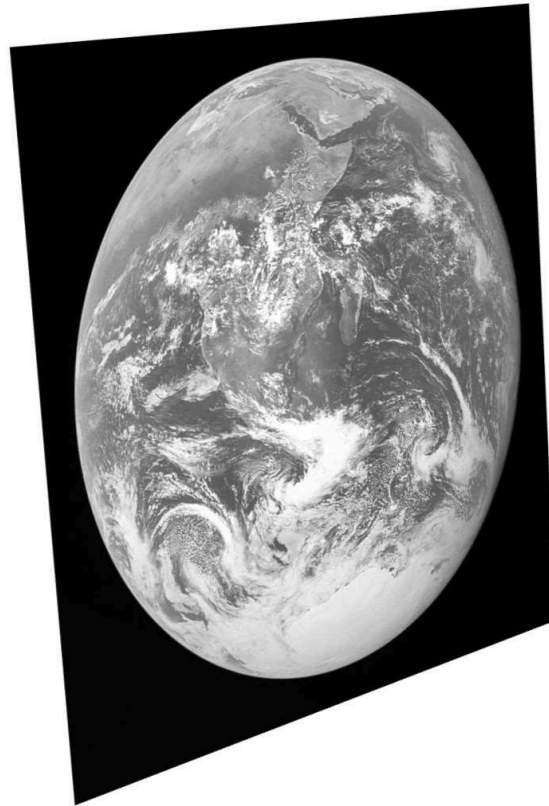
Continuous signal



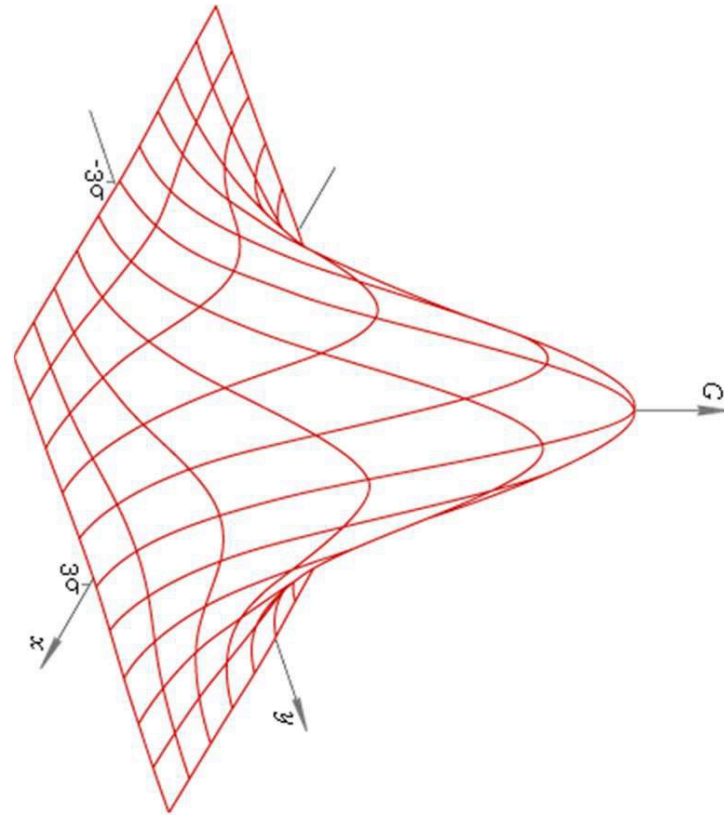
Camera

The imaging pipeline converts a continuous physical signal into discrete sensor measurements.

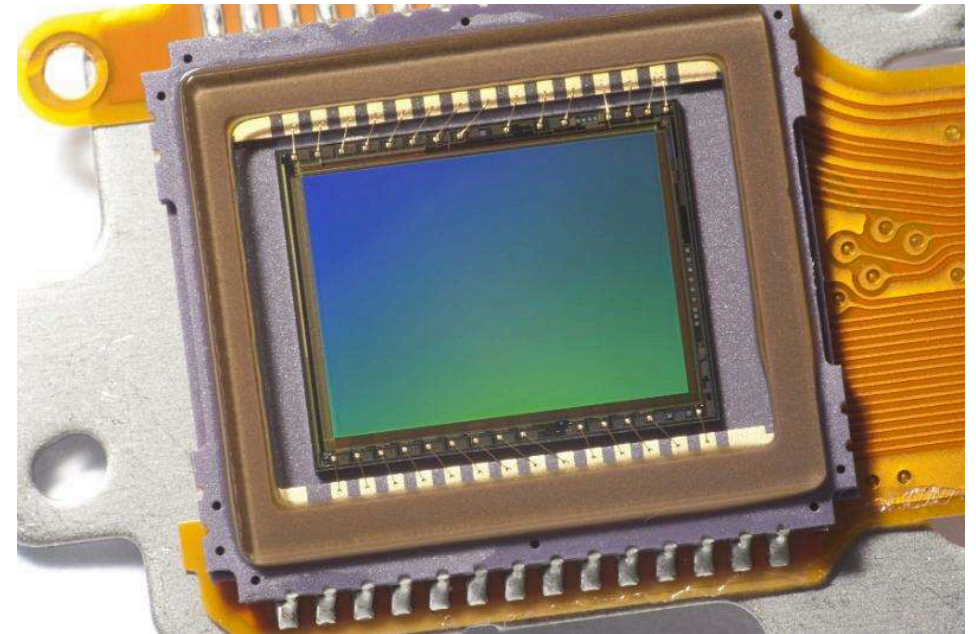
Sampling in Practice: Digital Image



Scene / image plane



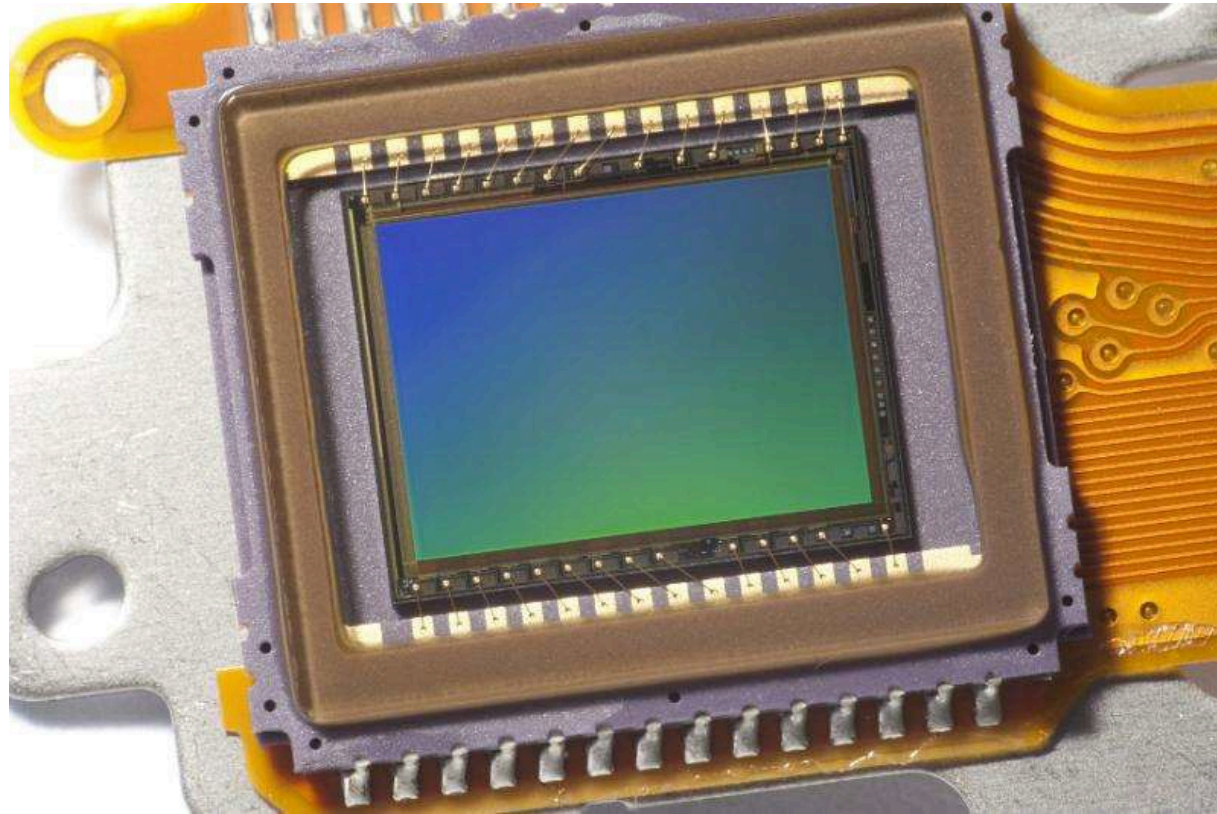
Light signal



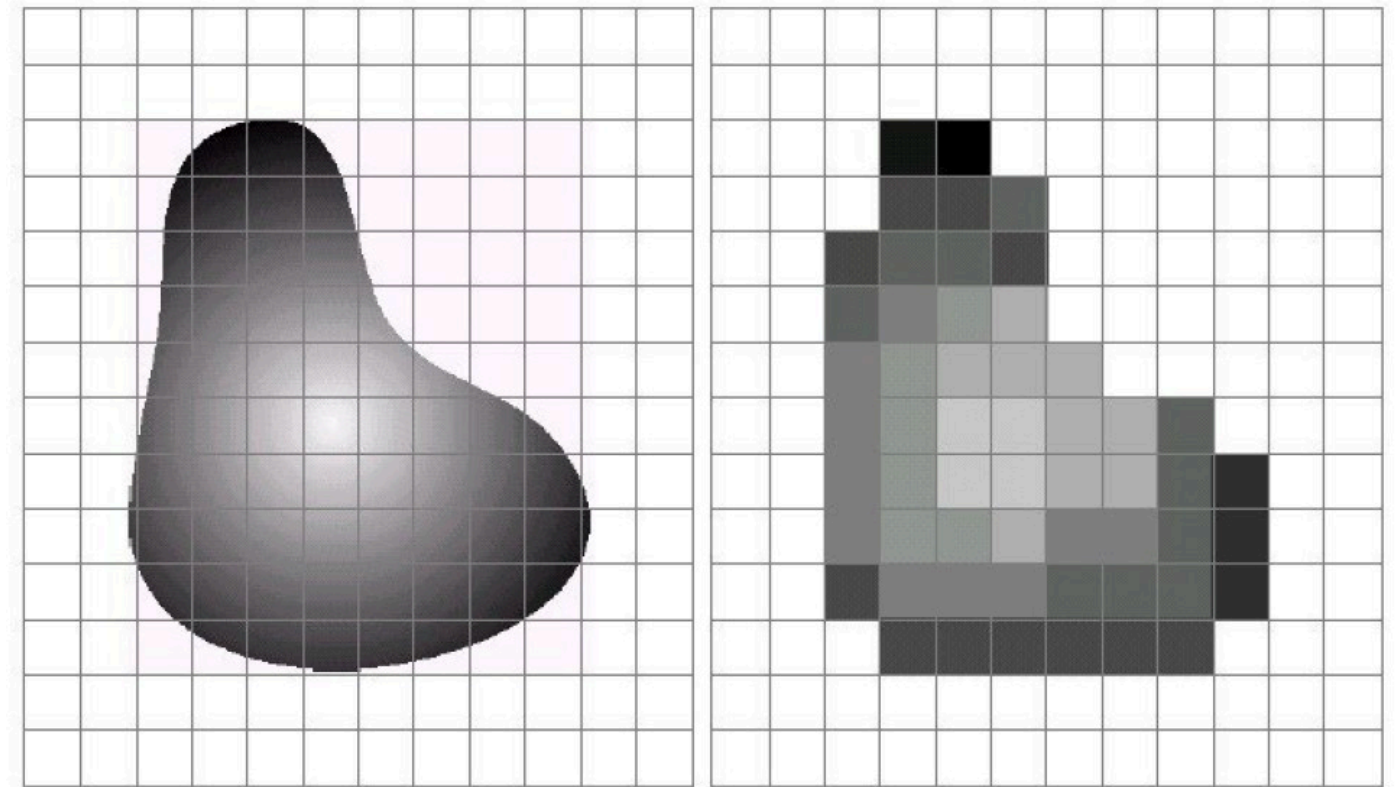
CMOS / CCD sensor

Digital cameras use sensor arrays to sample light and produce a grid of pixel values.

Sensor Array



Camera / sensor hardware



CMOS sensor array

Continuous light is **sampled** and **quantized** by the 2D sensor array (pixels) to produce a digital image.

Elements of a Digital Image

A pixel is a **picture element**.

For a grayscale image:

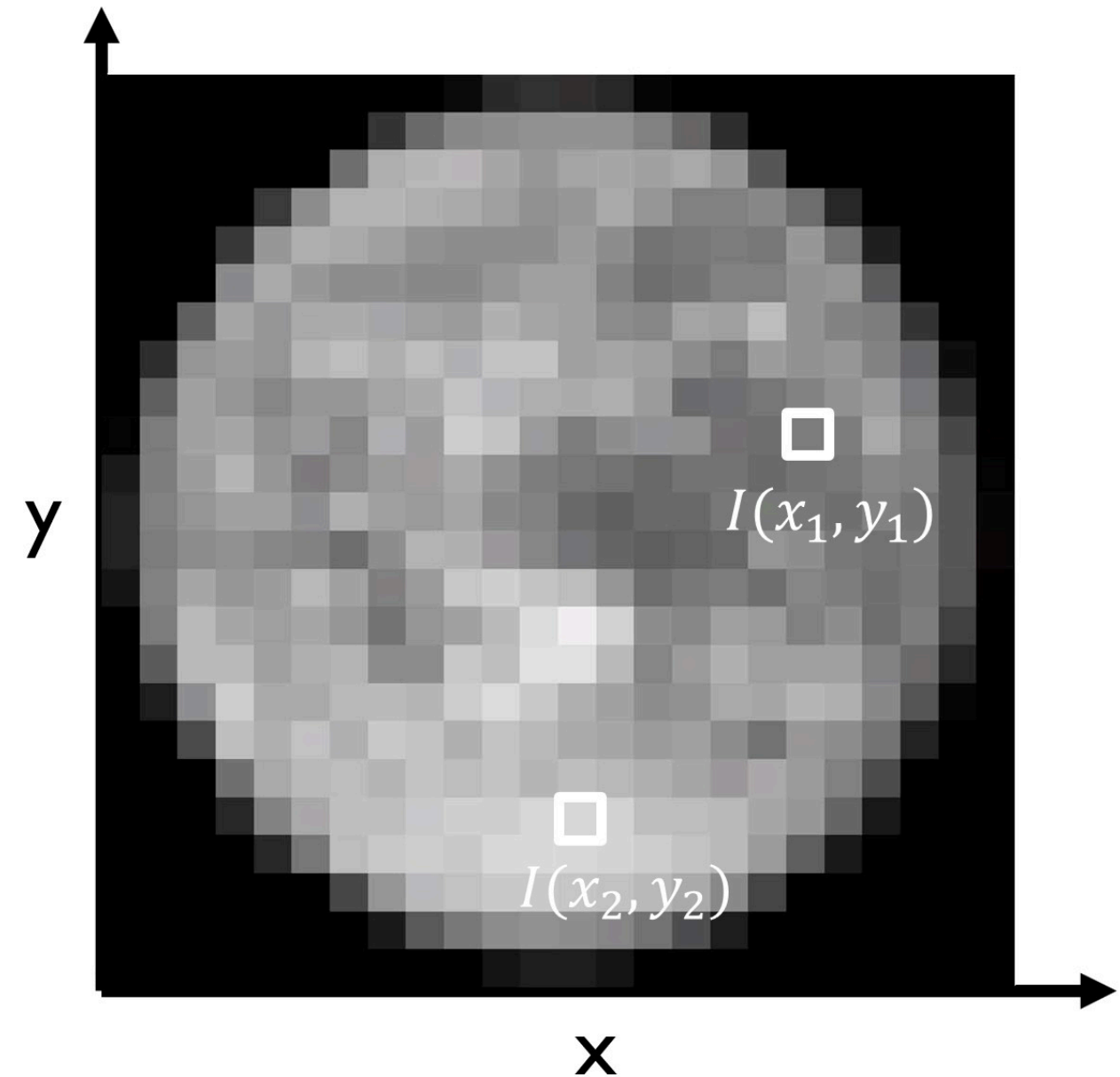
$$I(x, y) = \text{intensity at image location}(x, y)$$

- x : horizontal coordinate
- y : vertical coordinate

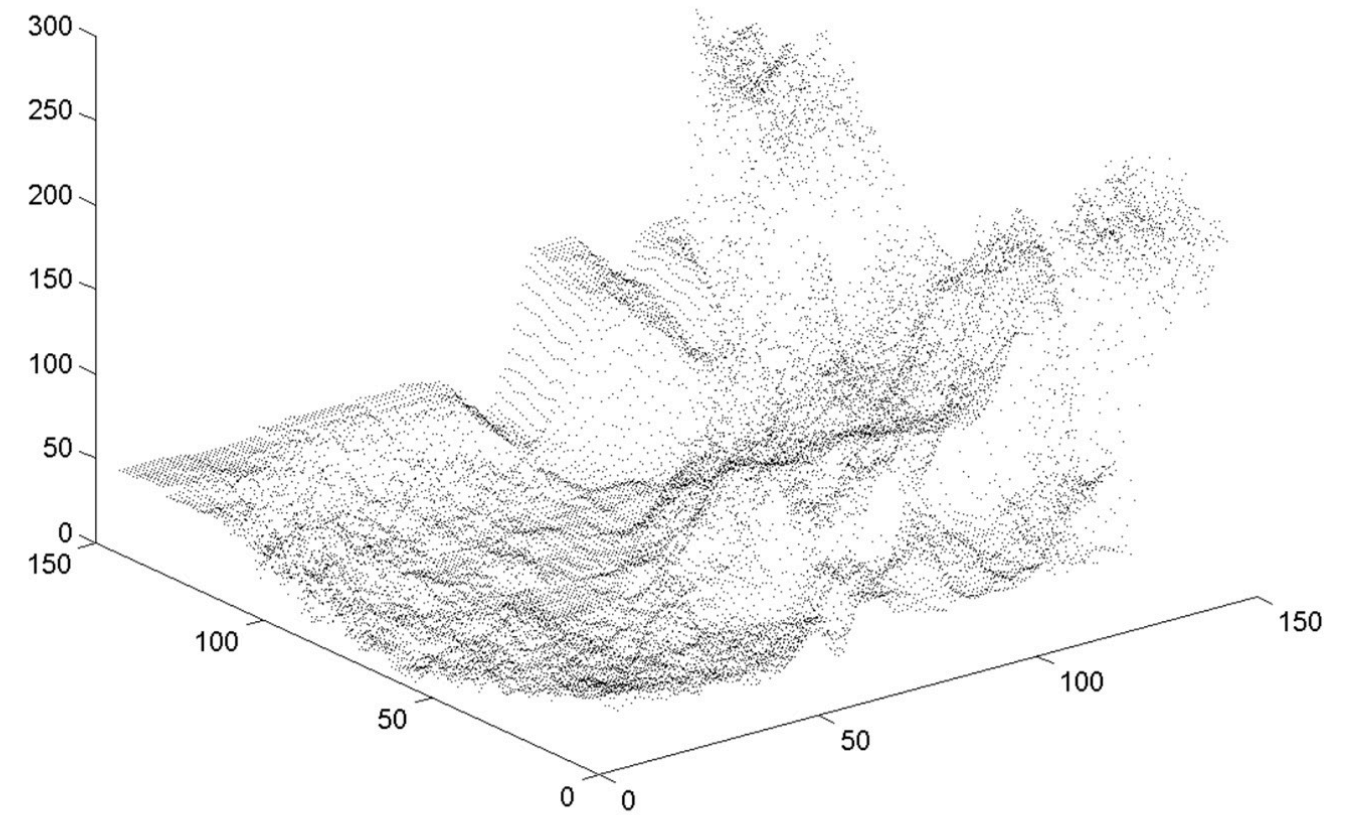
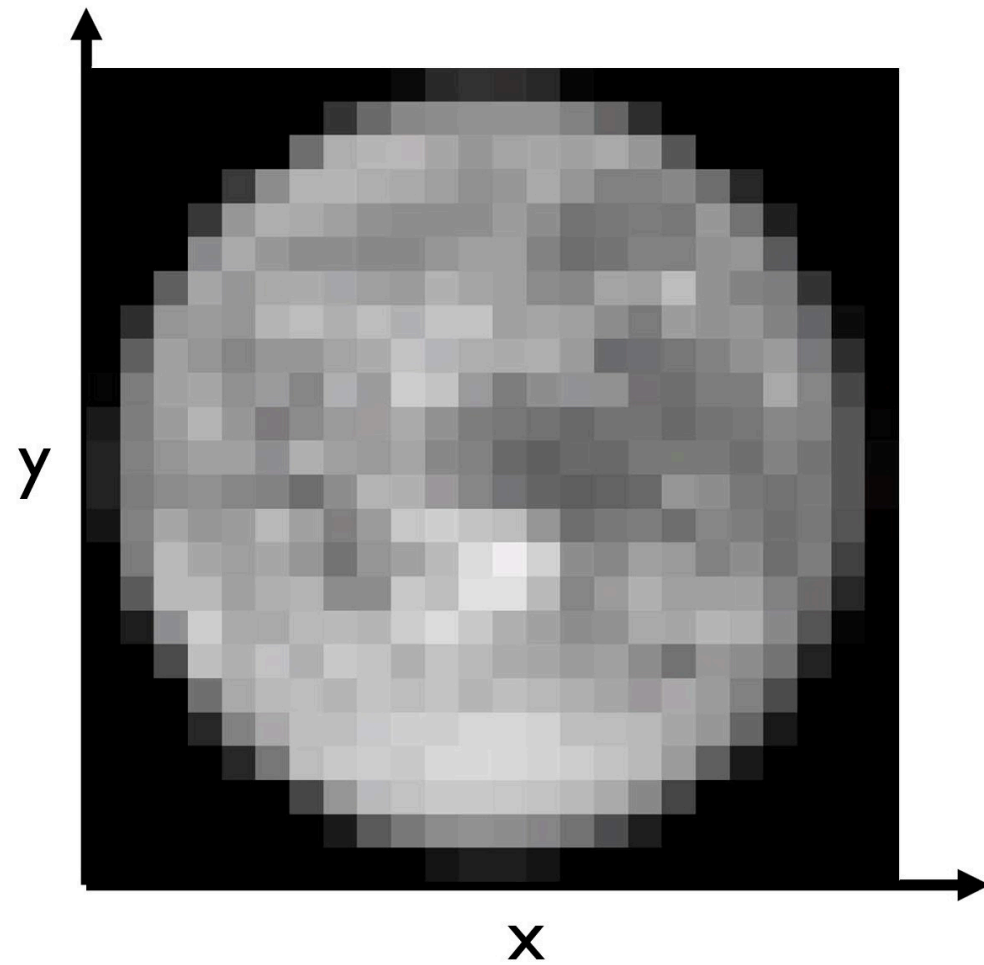
Example:

```
1 I(x1, y1) = 101  
2 I(x2, y2) = 191
```

Different pixel locations can have different intensity values.



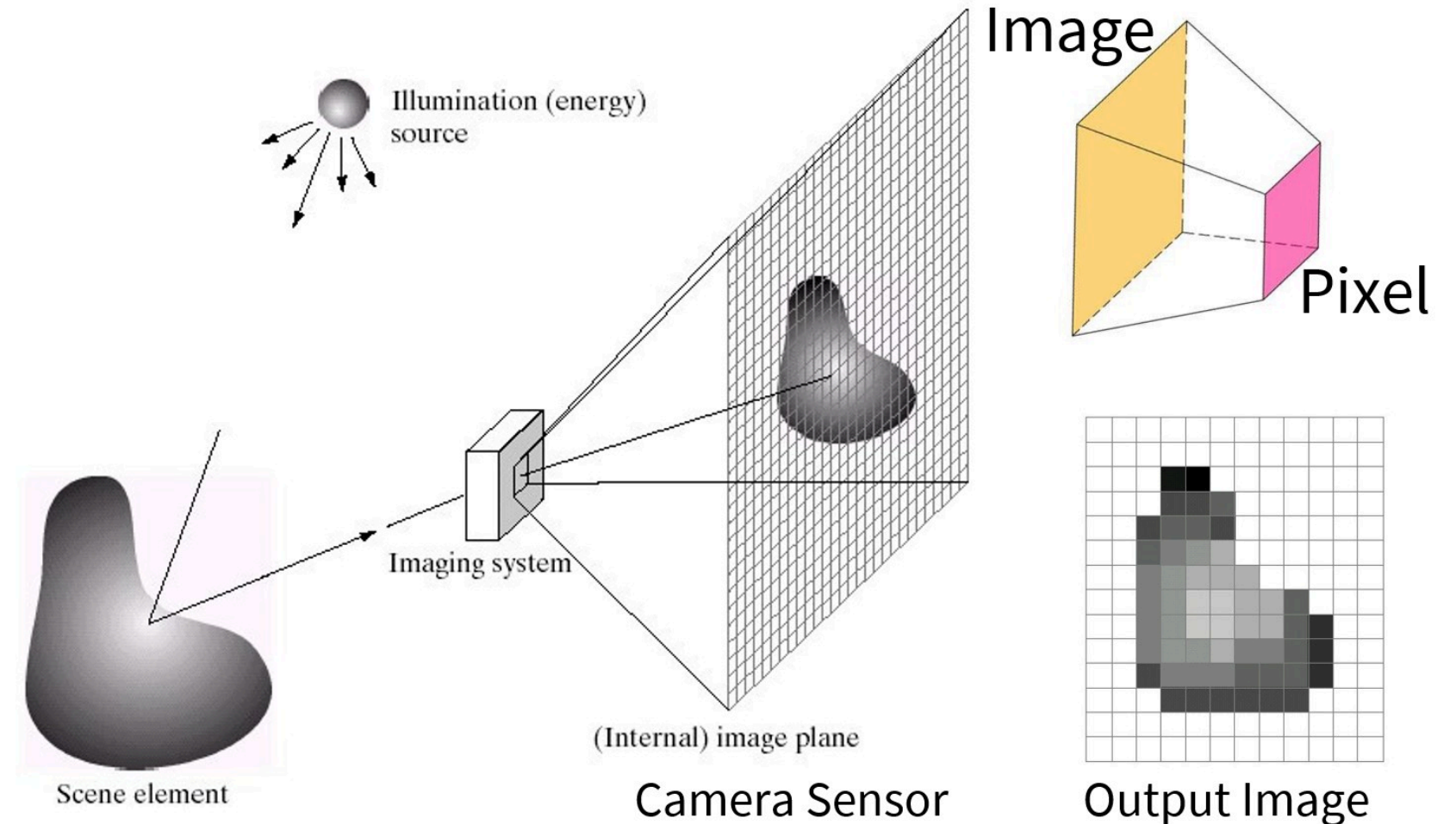
Digital Image is a 2D Signal



A digital image can be interpreted as a 2D function whose value is **brightness** or **intensity**.

Light Integration Over a Pixel Region

- A single pixel does not measure an infinitely thin ray.
- It measures light integrated over a small region of the sensor and the corresponding cone/frustum of incoming light.

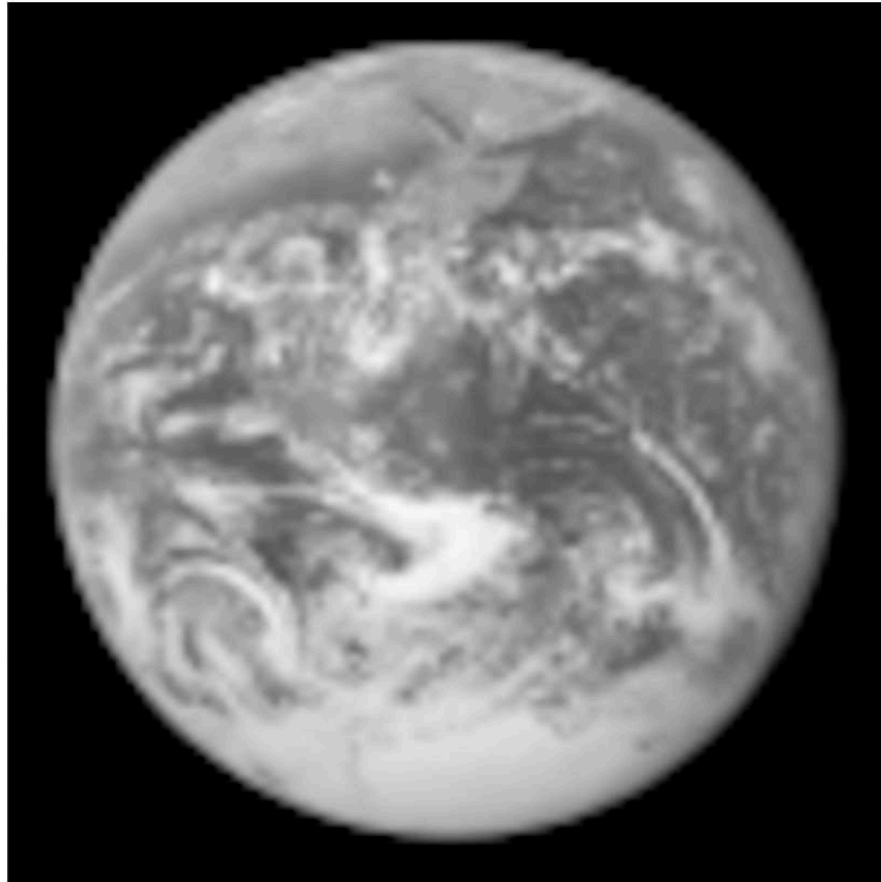


i Note

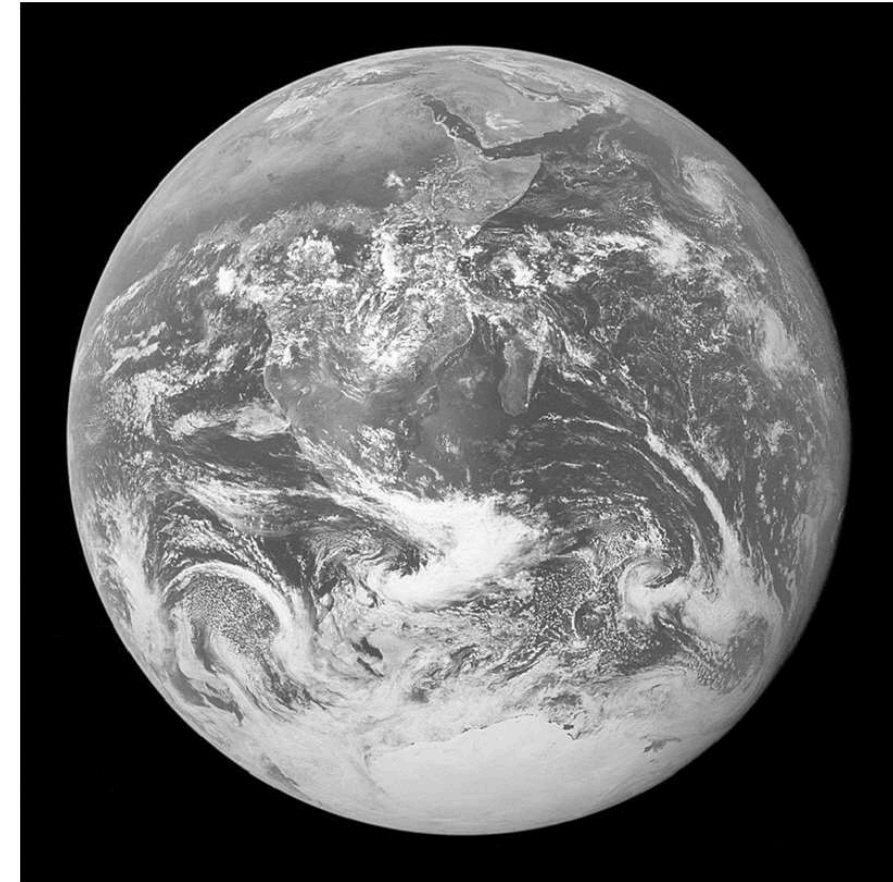
This is one reason why cameras blur, average, and alias details that are smaller than a pixel can represent.

Resolution: Geometric vs. Spatial

Both images can have the same number of pixels, but the amount of **scene detail** represented can be different.



Low geometric resolution

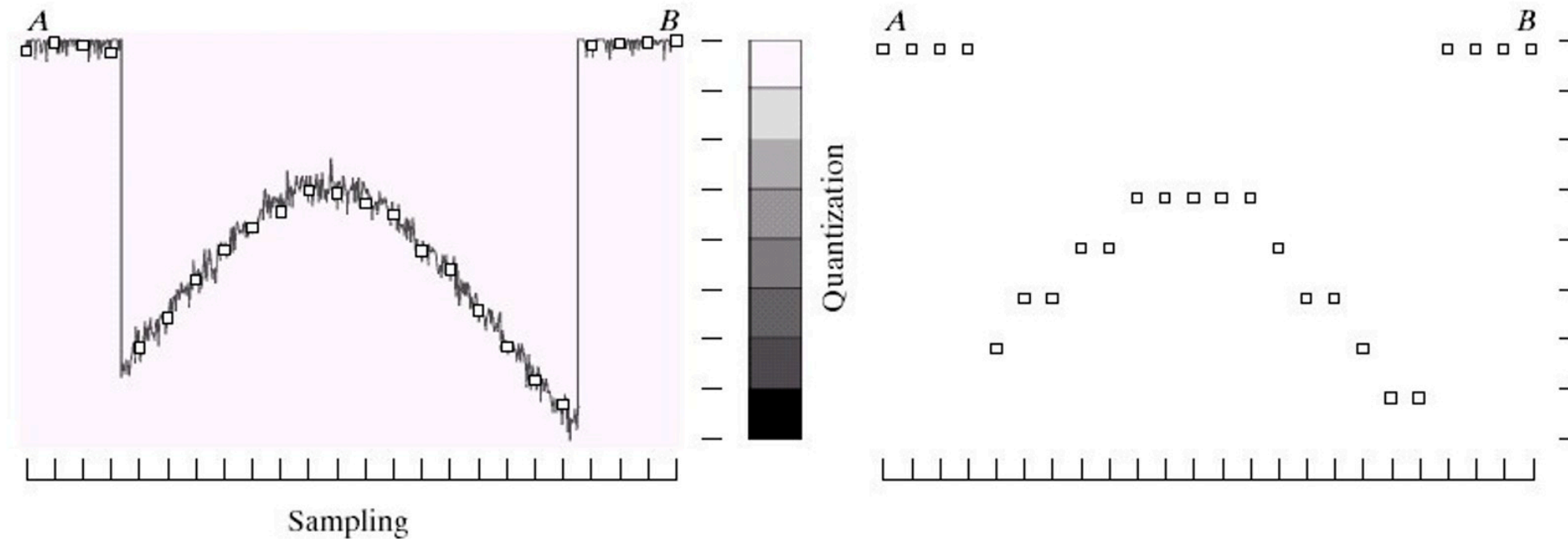


High geometric resolution

i Note

Spatial resolution counts pixels. Geometric resolution is how accurately those pixels represent the positions of objects in the real world; or how sharp the image is. Geometric resolution is limited by physics, optics, and the environment (see Rayleigh criterion).

Quantization

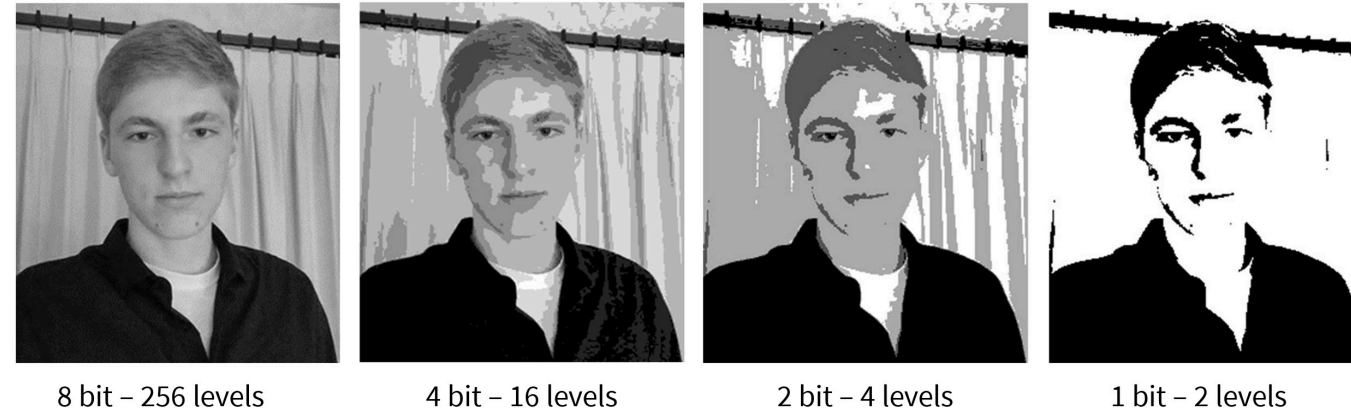


- Sampling chooses **where** to measure a signal.
- Quantization chooses **which values** can be represented.
- For example, an 3 bits can represent:

$$2^3 = 8$$

possible intensity levels in a grayscale image.

Quantization Effects — Radiometric Resolution

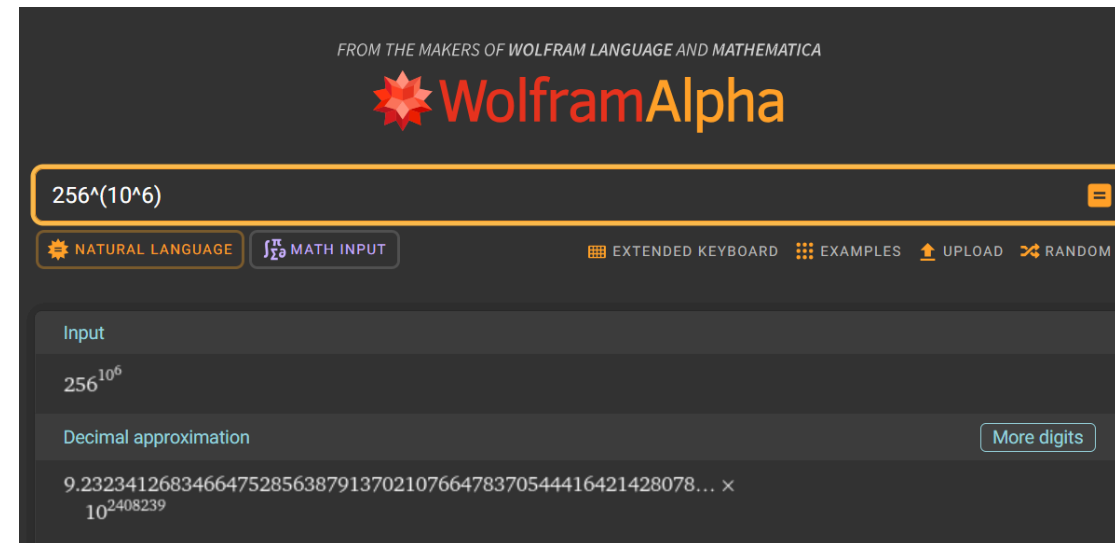


- Radiometric resolution is the number of distinct intensity levels available.

Bit depth	Number of levels
1 bit	2 levels
2 bit	4 levels
4 bit	16 levels
8 bit	256 levels

- Higher bit depth can represent more subtle intensity differences.
- In photography, this is closely related to dynamic range.

Dimensionality of an Image



- An image of size 1000×1000 with 8-bit quantization per pixel has:

$$256^{1000 \times 1000} \approx 10^{2,408,239}$$

possible intensity configurations.

- This is an enormous high-dimensional space – the known universe has only about 10^{80} atoms!
- Computer vision works in this extremely high dimensional space – but real images are not arbitrary arrays and they occupy a small, structured subset of this space.

Images in Python: Grayscale Arrays

For an $N \times M$ grayscale image `im`:

```
1 im[0, 0]      # top-left pixel value
2 im[y, x]      # y rows down, x columns right
3 im[N-1, M-1]  # bottom-right pixel value
```

0.92	0.93	0.94	0.97	0.62	0.37	0.85	0.97	0.93	0.92	0.99
0.95	0.89	0.82	0.89	0.56	0.31	0.75	0.92	0.81	0.95	0.91
0.89	0.72	0.51	0.55	0.51	0.42	0.57	0.41	0.49	0.91	0.92
0.96	0.95	0.88	0.94	0.56	0.46	0.91	0.87	0.90	0.97	0.95
0.71	0.81	0.81	0.87	0.57	0.37	0.80	0.88	0.89	0.79	0.85
0.49	0.62	0.60	0.58	0.50	0.60	0.58	0.50	0.61	0.45	0.33
0.86	0.84	0.74	0.58	0.51	0.39	0.73	0.92	0.91	0.49	0.74
0.96	0.67	0.54	0.85	0.48	0.37	0.88	0.90	0.94	0.82	0.93
0.69	0.49	0.56	0.66	0.43	0.42	0.77	0.73	0.71	0.90	0.99
0.79	0.73	0.90	0.67	0.33	0.61	0.69	0.79	0.73	0.93	0.97
0.91	0.94	0.89	0.49	0.41	0.78	0.78	0.77	0.89	0.99	0.93

⚠ Important

In NumPy, image indexing is usually **row first, column second**: `im[y, x]`, not `im[x, y]`.

Grayscale Intensity



A grayscale image is a **matrix** that stores one intensity value per pixel.

Color Images

- A color image stores multiple intensity values per pixel.
- For RGB images, each pixel has three channels:
 - Red intensity
 - Green intensity
 - Blue intensity

1 `image shape: height × width × 3`



i Note

Different libraries may use different channel order conventions. OpenCV commonly uses BGR; Matplotlib commonly expects RGB.

Images in Python: Color Arrays

For an $N \times M$ color image `im` with three channels:

```
1 im[0, 0, 0]      # top-left pixel, first channel
2 im[y, x, 1]      # pixel at row y, column x, second channel
3 im[N-1, M-1, 2]  # bottom-right pixel, third channel
```

Typical shape:

```
1 im.shape # (height, width, channels)
```

For an RGB image:

```
1 channel 0 = red
2 channel 1 = green
3 channel 2 = blue
```

For OpenCV-loaded images:

```
1 channel 0 = blue
2 channel 1 = green
3 channel 2 = red
```

Images in Python: Data Types

Take care of image data types.

```
1 # uint8 image: values 0 to 255
2 im = cv2.imread("file.jpg")
3
4 # convert to float32, still values 0 to 255
5 im_float = im.astype(np.float32)
6
7 # normalize to 0 to 1
8 im_norm = im.astype(np.float32) / 255.0
```

⚠ Important

Many image-processing bugs come from mixing `uint8`, `float32`, and different value ranges.

Random Images in Python

```
1 from numpy import random as r
2
3 I = r.rand(256, 256)
```

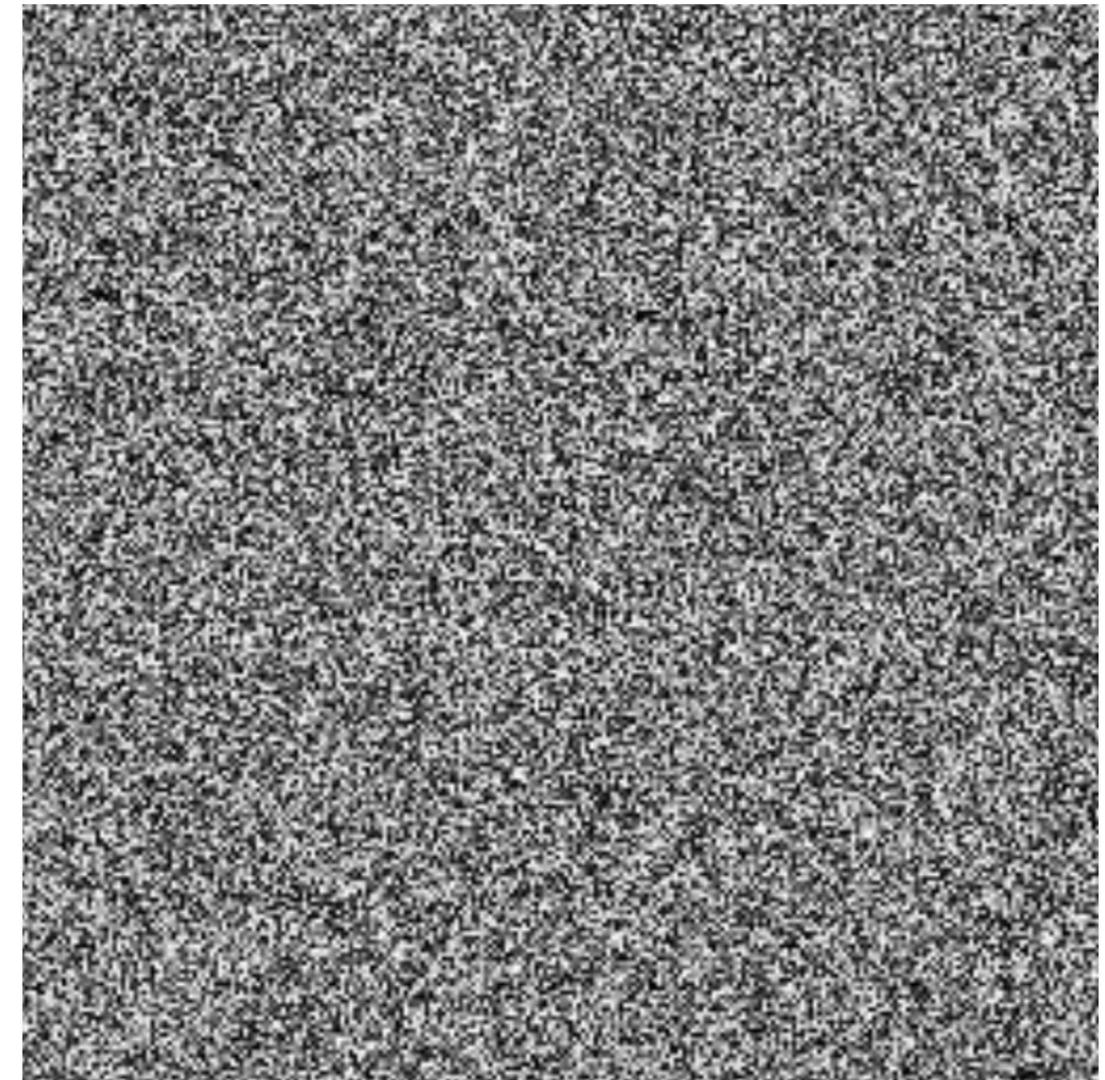
`r.rand(256, 256)` creates a 256 by 256 array of random floating-point values in the range $[0, 1)$.

Questions:

- What is `I`?
- What does it look like?
- Which values does it contain?
- How many values are there?

Displaying an Image in Python

```
1 from matplotlib import pyplot as plt
2 from numpy import random as r
3
4 I = r.rand(256, 256)
5
6 plt.imshow(I, cmap="gray")
7 plt.axis("off")
8 plt.show()
```



Is this an image?

Coding in This Course

We will use:

- **Python** for programming
- **Conda** for managing the course environment
- **NumPy** for numerical arrays
- **OpenCV** for image processing and computer vision
- **PyTorch** for deep learning topics
- **VS Code** for editing, debugging, and notebooks

⚠ Important

Do not use GenAI to solve homework for you. If you copy-paste code without understanding it, you will not learn how to code.

Be kind and supportive: students come from different backgrounds and have different levels of Python experience.

Python / Tool Refresher

Course tutorials are available here:

<https://ariarobotics.github.io/cv/tutorials/>

Review as needed:

- **WSL + Ubuntu** — recommended Linux environment for Windows users
- **Git & GitHub** — cloning, committing, pulling, pushing, and GitHub Classroom
- **Python & Conda** — installing Conda and creating the course environment
- **Visual Studio Code** — Python interpreter, notebooks, and debugging
- **NumPy** — arrays, indexing, vectorization, matrices, and image-related operations



Tip

These tutorials are reference material. Use them when you need help with the tools so class time can focus on computer vision.

Recommended Development Setup

Windows

Recommended workflow:

```
1 Windows
2   ↓
3 WSL 2 + Ubuntu
4   ↓
5 Miniforge / Conda
6   ↓
7 cv environment
8   ↓
9 VS Code + WSL extension
```

Keep course Python environments and projects **inside WSL**.

macOS / Linux

Recommended workflow:

```
1 macOS / Linux
2   ↓
3 Terminal
4   ↓
5 Miniforge / Conda
6   ↓
7 cv environment
8   ↓
9 VS Code
```

No WSL is needed.

Before We Create the Environment

Open a terminal and check the tools you already have.

Git

```
1 git --version
```

Conda

```
1 conda --version
```

Existing Conda environments

```
1 conda env list
```

Windows users working in WSL can also check:

```
1 uname -a
```



If `conda` is not installed yet, open the **Python & Conda** tutorial and follow the Miniforge installation section.

The Course Environment

We use a dedicated Conda environment named:

```
1 cv
```

The environment file is maintained in the course repository:

<https://github.com/ariarobotics/cv/blob/main/code/cv-environment.yml>

It defines the software stack used by the course, including:

- Python
- NumPy / SciPy
- OpenCV
- PyTorch / TorchVision
- Matplotlib
- scikit-image / scikit-learn
- Jupyter and VS Code notebook support

Do not manually assemble your own version of the environment.

Get the Environment File

Option 1 — Clone the course repository

```
1 cd ~/cv_projects
2 git clone https://github.com/ariarobotics/cv.git
3 cd cv/code
```

You should now see:

```
1 ls
```

including:

```
1 cv-environment.yml
```

Option 2 — Download only the environment file

```
1 curl -L -o cv-environment.yml \
2 https://raw.githubusercontent.com/ariarobotics/cv/main/code/cv-environment.yml
```

Create the **cv** Environment

From the directory containing `cv-environment.yml`:

```
1 conda env create -f cv-environment.yml
```

This may take several minutes.

The environment file already specifies the name:

```
1 cv
```

When installation finishes:

```
1 conda activate cv
```

Your terminal prompt should now begin with something similar to:

```
1 (cv)
```

Important

Whenever you run course Python code, make sure the **cv** environment is active.

Verify the Environment

With `cv` activated:

```
1 python --version
```

Check which Python is being used:

```
1 which python
```

Then verify the main packages:

```
1 python -c "import numpy; print('NumPy:', numpy.__version__)"
2 python -c "import cv2; print('OpenCV:', cv2.__version__)"
3 python -c "import torch; print('PyTorch:', torch.__version__)"
4 python -c "import torchvision; print('TorchVision:', torchvision.__version__)"
```

If all four commands work, the core environment is ready.

Configure VS Code

Open the project folder from your terminal:

```
1 code .
```

Windows + WSL users should see **WSL: Ubuntu** in the VS Code status area.

Then:

1. Open the Command Palette: **Ctrl+Shift+P**
2. Search for **Python: Select Interpreter**
3. Select the interpreter associated with **cv**

For notebooks, use **Select Kernel** and choose the same **cv** environment.

Verify from Python:

```
1 import sys
2 print(sys.executable)
```

The path should point into the **cv** Conda environment.

Our First Computer Vision Program

Create a file named:

```
1 first_cv.py
```

Start by importing the libraries:

```
1 from pathlib import Path
2
3 import cv2
4 import matplotlib.pyplot as plt
5 import numpy as np
```

Create a folder for output files:

```
1 output_dir = Path("outputs")
2 output_dir.mkdir(exist_ok=True)
```

Create and Save a Simple Image

For today's example, we will create a small synthetic image so everyone has the same input.

```
1 height = 360
2 width = 640
3
4 image = np.full(
5     (height, width, 3),
6     235,
7     dtype=np.uint8,
8 )
9
10 # OpenCV uses BGR color ordering.
11 cv2.rectangle(
12     image, (70, 80), (260, 280),
13     (255, 100, 0), thickness=-1
14 )
15
16 cv2.circle(
17     image, (470, 180), 90,
```

We now have a real image file on disk that we can load with OpenCV.

Load and Inspect the Image

Load the image:

```
1 image_bgr = cv2.imread(  
2     str(output_dir / "sample_image.png")  
3 )
```

Inspect it:

```
1 print("Shape:", image_bgr.shape)  
2 print("Data type:", image_bgr.dtype)
```

You should see something similar to:

```
1 Shape: (360, 640, 3)  
2 Data type: uint8
```

An image is fundamentally a **NumPy array**.

Display the Image

OpenCV loads color images in **BGR** order.

Matplotlib expects **RGB**, so convert before displaying:

```
1 image_rgb = cv2.cvtColor(  
2     image_bgr,  
3     cv2.COLOR_BGR2RGB  
4 )  
5  
6 plt.imshow(image_rgb)  
7 plt.axis("off")  
8 plt.show()
```

A useful rule:

```
1 OpenCV processing → usually BGR  
2 Matplotlib display → RGB
```

Note

Using Matplotlib is also convenient in notebooks and avoids problems that can occur with `cv2.imshow()` in some WSL setups.

A First Image-Processing Operation

Convert the image to grayscale:

```
1 gray = cv2.cvtColor(  
2     image_bgr,  
3     cv2.COLOR_BGR2GRAY  
4 )
```

Detect edges:

```
1 edges = cv2.Canny(  
2     gray,  
3     100,  
4     200  
5 )
```

Display the result:

```
1 plt.imshow(edges, cmap="gray")  
2 plt.axis("off")  
3 plt.show()
```

What Just Happened?

In only a few lines, we used several ideas that will appear throughout the course:

- An image is represented as a **NumPy array**
- Image dimensions are described by `.shape`
- Pixel values have a data type such as `uint8`
- OpenCV can **load, save, transform, and analyze images**
- Color-channel conventions matter: **BGR vs. RGB**
- Computer vision pipelines are often sequences of simple operations

```
1 image
2   ↓
3 load
4   ↓
5 represent as an array
6   ↓
7 transform / analyze
8   ↓
9 display or use the result
```

Common Setup Problems

conda: command not found

Restart the terminal or reload your shell:

```
1 source ~/.bashrc
```

Python imports fail

First check:

```
1 conda activate cv
2 which python
```

VS Code runs a different Python

Use:

```
1 Python: Select Interpreter
```

and select **cv**.

Windows paths and Linux paths are getting mixed

If you are using WSL, keep the course environment and projects inside the WSL/Linux filesystem.

After Class

Please go through any tutorials that cover tools you are not yet comfortable with:

<https://ariarobotics.github.io/cv/tutorials/>

In particular, make sure you can:

- Activate the `cv` environment
- Use basic Git commands
- Open a project in VS Code
- Run Python scripts and Jupyter notebooks
- Work with basic NumPy arrays

You do **not** need to memorize every Git, Conda, or NumPy command. The tutorials are there as references throughout the semester.